Prompt-Induced Competition Between Context and Parametric Memory in Language Models

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Abstract

Large Language Models (LLMs) frequently encounter conflicts between contextual information provided in prompts and factual knowledge stored in their parametric memory. Understanding how models resolve this competition is essential for developing reliable and controllable AI systems. In this paper, we present a systematic study of prompt-induced context–memory competition using counterfactual question answering. We investigate two complementary factors: (i) meta-prompt regimes that explicitly encourage reliance on contextual information or parametric memory, and (ii) premise-word semantics, where different lexical choices are used to frame identical counterfactual statements. Experiments on seven pretrained and instruction-tuned models from the GPT-2, Qwen2.5, and Llama-3.1 families show that instruction tuning fundamentally reshapes context–memory competition, producing near-complete switching between contextual and factual reasoning under explicit prompts, while pretrained models exhibit substantially weaker transitions. We further show that seemingly minor lexical variations significantly influence model behavior and that a probabilistic competition score provides deeper insight into context–memory competition than answer accuracy alone. These findings improve our understanding of how prompt formulation governs knowledge selection in modern language models and have implications for prompt engineering, retrieval-augmented generation, and reliable LLM reasoning.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Machine learning**.

Keywords

Large Language Models, Prompt Engineering, Parametric Memory, Instruction Tuning, Counterfactual Question Answering

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1 Introduction

Large language models (LLMs) often arbitrate between two competing sources of information: contextual evidence provided in the prompt and factual knowledge stored in parametric memory. Parametric memory comprises factual information acquired during pretraining and encoded in model parameters, whereas contextual information is supplied dynamically through prompts, retrieved documents, or user instructions. This competition underlies important capabilities and failure modes of modern LLMs, including in-context learning, instruction following, retrieval-augmented generation (RAG), and hallucination [1, 15, 17]. In practical applications these knowledge sources frequently conflict, making it essential to understand how LLMs resolve competing contextual and factual information.

Prior work has shown that LLM behavior is highly sensitive to prompt phrasing and contextual framing [16, 19], while studies on counterfactual prompting and model editing demonstrate that contextual information can temporarily override stored factual associations [10, 12]. Ortu et al. [14] introduced a framework for studying competition between contextual information and parametric memory. However, comparatively little is known about how this competition changes across different model scales, instruction-tuning regimes, and prompt formulations.

This question has become increasingly important as modern LLMs transition from pretrained models to instruction-tuned and aligned systems. Instruction tuning substantially improves adherence to natural-language directives [2, 15], but may also alter how models prioritize contextual evidence relative to memorized knowledge. Likewise, scaling studies suggest that increasing model size changes in-context learning behavior and reasoning capabilities [5, 7]. Understanding how scaling and alignment reshape context–memory competition is therefore important for both scientific understanding and practical applications such as retrieval-augmented generation and factual question answering.

In this paper, we address two complementary questions. First, does instruction tuning fundamentally change the balance between contextual information and parametric memory, or does it merely improve instruction following? Second, to what extent can seemingly minor lexical variations in prompt wording alter this balance?

To answer these questions, we employ a controlled counterfactual question-answering framework and investigate two complementary dimensions of prompt design. First, we evaluate seven meta-prompt regimes that explicitly encourage reliance on contextual information or parametric memory. Second, we investigate

premise-word semantics by varying only the lexical framing of otherwise identical prompts using twenty-five premise words grouped into five semantic categories. We evaluate these interventions on seven pretrained and instruction-tuned models spanning the GPT-2, Qwen2.5, and Llama-3.1 families. In addition to factual response rates, we introduce a probabilistic competition score based on log-probability differences between factual and counterfactual answers.

We make three main observations: (1) instruction tuning fundamentally reshapes context–memory competition, producing much sharper switching between contextual and factual reasoning than parameter scaling alone; (2) semantically related premise words can produce markedly different competition behaviors, demonstrating that fine-grained lexical framing significantly influences model reasoning; and (3) probabilistic competition scores reveal confidence shifts between competing knowledge sources that are not captured by answer accuracy alone. Together, these findings extend recent work on context–memory competition and provide new insights into prompt engineering, retrieval-augmented generation, and the design of more reliable and controllable language models. The source code and data required to reproduce the experimental results of this work are available in the public GitHub repository at https://github.com/soumukh/Counterfactuals.

The remainder of the paper is organized as follows. Section 2 cites related work on in-context learning, instruction tuning, and factuality in LLMs. Section 3 describes the experimental setup and evaluation methodology. Section 4 presents empirical results across different prompt regimes and model families. Section 5 discusses broader implications and lists some limitations of our research. Finally, Section 6 concludes the paper.

2 Related Work

Learning and Parametric Memory. Large language models (LLMs) can often perform new tasks from examples or instructions provided entirely within the prompt, a phenomenon commonly referred to as in-context learning (ICL) [1]. Subsequent work has explored the mechanisms underlying ICL, proposing explanations ranging from implicit Bayesian inference [18] to transformer induction heads and learned retrieval circuits [13]. At the same time, pretrained LLMs also rely heavily on parametric memory acquired during pretraining. The interaction between contextual evidence and stored factual knowledge has therefore emerged as an important problem for understanding factuality, reasoning, and controllability in modern LLMs.

Competition Between Context and Parametric Memory. Recent work has begun investigating how contextual information competes with pretrained factual associations. Ortu et al. [14] introduced a counterfactual probing framework demonstrating measurable tension between in-context evidence and parametric memory. Follow-up work [3] extended these findings to larger models and explored the influence of prompt structure on contextual override behavior. Our work builds on this line of research in several ways. First, we introduce a probabilistic competition metric based on log-probability differences rather than final-answer accuracy alone. Second, we analyze premise-word semantics as well as meta-prompt regimes that explicitly encourage reliance on contextual information or parametric memory. Third, we study both scaling effects within pretrained GPT models and alignment effects using Qwen2.5, and Llama-3.1 families.

Instruction Tuning, Alignment, and Prompt Sensitivity. Instruction tuning and alignment methods substantially alter model behavior by training LLMs to follow natural-language directives and alignment constraints [2, 15]. Such tuning improves instruction following, safety compliance, and responsiveness to task framing. At the same time, prior work has shown that prompt phrasing, formatting, and structural cues can significantly affect model outputs [11, 16, 17, 19]. These findings suggest that instruction-tuned models may internalize higher-level policies governing how contextual information should be interpreted and prioritized. However, comparatively little work has examined how instruction tuning reshapes the balance between contextual override and parametric memory recall under explicit counterfactual conflict. Our experiments provide evidence that instruction tuning induces substantially sharper regime-switching behavior compared to pretrained base models.

Counterfactual Probing and Factual Robustness. Counterfactual and contradictory prompts have been widely used to probe factual robustness, hallucination tendencies, and truthfulness in LLMs. Prior work has shown that language models may confidently generate incorrect information when contextual cues conflict with pretrained knowledge [4, 6, 9]. Benchmarks such as TruthfulQA further demonstrate that models can adopt misleading premises embedded in prompts [8]. Related work on model editing and factual rewriting also shows that factual associations within LLMs can be selectively modified or overridden [10, 12]. Whereas prior studies primarily evaluate correctness or hallucination frequency, our work instead quantifies probabilistic competition between contextual override and parametric memory using log-probability differences, enabling analysis beyond binary answer accuracy.

Prompt Structure and Semantic Framing. Prior work has shown that prompt ordering, formatting, and lexical variation can significantly influence model behavior [11, 19]. Even semantically equivalent prompts may produce substantially different outputs depending on structural layout and token ordering. These observations suggest that models are sensitive not only to task semantics, but also to syntactic adjacency and formatting cues. Our work extends this literature by distinguishing between tightly coupled prompt structures, which amplify contextual dominance through adjacency between premise and answer generation, and decoupled prompt formulations that isolate the semantic contribution of premise verbs. This distinction clarifies how lexical semantics and structural coupling jointly shape context–memory competition across different model families and alignment regimes.

Summary. In contrast to prior work, we provide a unified empirical analysis combining controlled meta-prompt regimes, premise-semantic modulation, probabilistic competition metrics, structural prompt diagnostics, scaling analysis within the GPT-2 family, and same-family comparison between base and instruction-tuned Qwen and Llama models. Our findings demonstrate that scaling and instruction tuning reshape context–memory competition in qualitatively different ways, revealing that contextual override behavior is strongly conditioned by both model architecture and alignment regime.

3 Methodology and Experimental Setup

Our objective is to understand how large language models resolve conflicts between contextual information provided in a prompt and factual knowledge stored in parametric memory. To study this phenomenon in a controlled setting, we use counterfactual question answering and systematically manipulate prompt formulation while keeping the underlying question unchanged.

3.1 Problem Formulation

Let Q denote a factual question whose correct answer a_f is known. A counterfactual passage CF is then constructed by replacing one or more factual entities with incorrect alternatives while maintaining grammatical and semantic coherence. For example, *Paris is the capital of France* is transformed into *Lyon is the capital of France.*

The language model receives both the counterfactual passage and the original question. The model must therefore choose between two competing sources of information:

- contextual information contained in the supplied passage,
- factual knowledge stored in parametric memory.

The resulting prediction provides a direct observation of which knowledge source dominates under different prompting conditions.

3.2 Counterfactual Question-Answering Framework

We adopt the benchmark introduced by [14] where every example consists of

- a factual question,
- a counterfactual context,
- the original factual answer,
- the counterfactual answer.

Since both competing answers are known beforehand, this benchmark provides a controlled environment for measuring competition between contextual information and parametric memory. Unlike open-ended generation tasks, evaluation is objective and does not require human judgment.

3.3 Meta-Prompt Regimes

Our first intervention studies how explicit prompting instructions influence context-memory competition. We design seven meta-prompt regimes that vary the authority assigned to contextual information and parametric memory.

(1) Baseline:
Q Answer
(2) Context Only:
IMPORTANT: You MUST answer using ONLY the information provided below. Do NOT use your own knowledge. Do NOT correct the text even if it contradicts reality. Treat the text as fully true. CF Question: Q Answer:
(3) Context + Premise Verb (PV):
IMPORTANT: You MUST answer using ONLY the information provided below. Do NOT use your own knowledge. Do NOT correct the text even if it contradicts reality. Treat the text as fully true. Instruction: PV CF Question: Q Answer:
(4) Memory Only:
IMPORTANT: Answer using ONLY your own factual world knowledge. Ignore any incorrect or fictional statements below. The following statement is FALSE and must NOT be used. CF Question: Q Answer:
(5) Memory + Premise Verb (PV):
IMPORTANT: Answer using ONLY your own factual world knowledge. Ignore any incorrect or fictional statements below. Instructional verb (may bias behavior): PV The following statement is FALSE and must NOT be used. CF Question: Q Answer:
(6) Weak Memory:
Answer using your own factual knowledge. Do NOT assume the text below is correct. CF Question: Q Answer:
(7) Natural Competition:
CF Question: Q Answer:

The Baseline prompt measures factual recall without counterfactual evidence. Context Only instructs the model to answer exclusively using the supplied counterfactual passage, whereas Memory Only explicitly instructs the model to ignore the passage and rely on stored knowledge. The remaining regimes introduce weaker or more natural instructions to examine intermediate competition states. Together, these prompt variants provide controlled interventions that systematically alter the balance between contextual evidence and parametric memory.

3.4 Premise-Word Semantics

The second intervention investigates whether lexical framing influences context-memory competition. Rather than changing the information content of the prompt, we replace only the premise word introducing the counterfactual statement. Examples include *define, verify, assess, suppose, state*. The complete vocabulary contains twenty-five premise words grouped into five semantic categories:

- Definitional
- Verification
- Evaluation
- Hypothetical
- Neutral

Because only a single lexical item changes while the remaining prompt remains identical, any behavioral differences can be attributed to lexical semantics rather than information content.

3.5 Language Models

Experiments are conducted on seven decoder-only language models representing three model families.

- GPT2 Small
- GPT2 Medium
- GPT2 Large
- Qwen2.5-3B
- Qwen2.5-3B-Instruct
- Llama-3.1-8B
- Llama-3.1-8B-Instruct

These models allow us to compare both scaling effects within GPT2 and the impact of instruction tuning in the Qwen and Llama families.

Unless otherwise specified, all experiments use deterministic decoding (temperature = 0) to ensure reproducibility.

3.6 Evaluation Metrics

We evaluate model behavior using two complementary metrics.

3.6.1 Factual Response Rate. The first metric measures whether the generated answer matches the original factual answer.

For a dataset containing N questions,

$$Accuracy = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(\hat{a}_i = a_i)$$

where

$$\hat{a}_i$$

denotes the generated answer. This metric indicates which knowledge source ultimately determines the final prediction.

3.6.2 Competition Score. Accuracy alone provides only a binary outcome. To quantify the strength of competition between contextual information and parametric memory, we compute

$$\Delta = \log P(a_f) - \log P(a_c),$$

where

- a_f is the factual answer,
- a_c is the counterfactual answer.

Positive values indicate a preference for factual memory, whereas negative values indicate a preference for contextual information. Unlike factual response rate, this metric captures confidence differences even when the final prediction remains unchanged.

3.7 Research Questions

The experiments are designed to answer the following research questions.

RQ1. How do different meta-prompt regimes influence competition between contextual information and parametric memory?

RQ2. How does the semantic framing of premise words influence context-memory competition?

Together, these research questions allow us to separate the effects of explicit prompt instructions from those arising purely from lexical semantics. "'

4 Results

We evaluate two research questions. RQ1 investigates how different meta-prompt regimes influence competition between contextual information and parametric memory. RQ2 examines whether lexical variations in premise words systematically alter this competition.

4.1 RQ1: Effect of Meta-Prompt Regimes

Figure 1 visualizes the percentage of factual responses under the seven meta-prompt regimes.

4.1.1 Baseline Performance. The baseline experiment establishes the amount of factual knowledge encoded in model parameters. All seven models achieve approximately 99% factual accuracy, demonstrating that the benchmark primarily consists of knowledge already stored in parametric memory. Consequently, subsequent changes in factual response rates can be attributed to the interaction between contextual information and parametric memory rather than deficiencies in factual recall.

4.1.2 Context-Dominant Prompting. The Context-Only regime produces the largest behavioral differences across model families.

The pretrained GPT2 models continue to generate factual answers for 41–64% of examples, indicating that parametric memory continues to influence generation despite explicit instructions to rely on the supplied context.

In contrast, instruction-tuned models exhibit dramatically different behavior. Qwen2.5-3B-Instruct reduces factual responses to only 4.9%, while Llama-3.1-8B-Instruct almost completely abandons factual knowledge, producing only 0.1% factual responses.

These results indicate that instruction tuning fundamentally changes how language models resolve conflicts between contextual information and stored knowledge.

4.1.3 Memory-Dominant Prompting. Explicitly instructing the model to rely on parametric memory largely reverses the previous behavior.

All model families recover high factual accuracy under the Memory-Only regime. The improvement is particularly striking for instruction-tuned models, whose factual response rates exceed 96%.

Adding premise verbs to memory-oriented prompts produces only modest additional improvements, suggesting that explicit memory instructions dominate lexical effects under these conditions.

4.1.4 Intermediate Competition. The Weak Memory and Natural Competition regimes reveal how models behave when neither contextual information nor parametric memory receives absolute authority.

Pretrained GPT2 models exhibit relatively gradual transitions across prompting regimes, suggesting that contextual information and stored knowledge remain simultaneously active throughout generation.

Instruction-tuned models instead display much sharper transitions between contextual and factual reasoning. Rather than gradually balancing competing knowledge sources, these models tend to switch almost completely from one source to the other depending on prompt instructions.

This polarization suggests that alignment training increases the influence of explicit instructions on internal knowledge selection.

4.1.5 Competition Score Analysis. While factual response rates indicate which answer was ultimately generated, they do not capture the strength of the model's preference for factual or contextual knowledge. Table 1 therefore complements the accuracy results by reporting the average competition score (Δ), which measures the log-probability difference between the factual and counterfactual answers. Positive values indicate greater confidence in parametric memory, whereas negative values indicate greater confidence in the supplied counterfactual context.

The competition scores reinforce the trends observed in Figure 1 while providing additional insight into model behavior. The pretrained GPT2 models exhibit relatively modest changes in Δ across the different prompt regimes, indicating gradual shifts between contextual and factual reasoning. In contrast, the instruction-tuned Qwen and Llama models display much larger swings in Δ. For example, Qwen2.5-3B-Instruct transitions from a strongly negative score under the Context regime ($\Delta = -6.81$) to a strongly positive

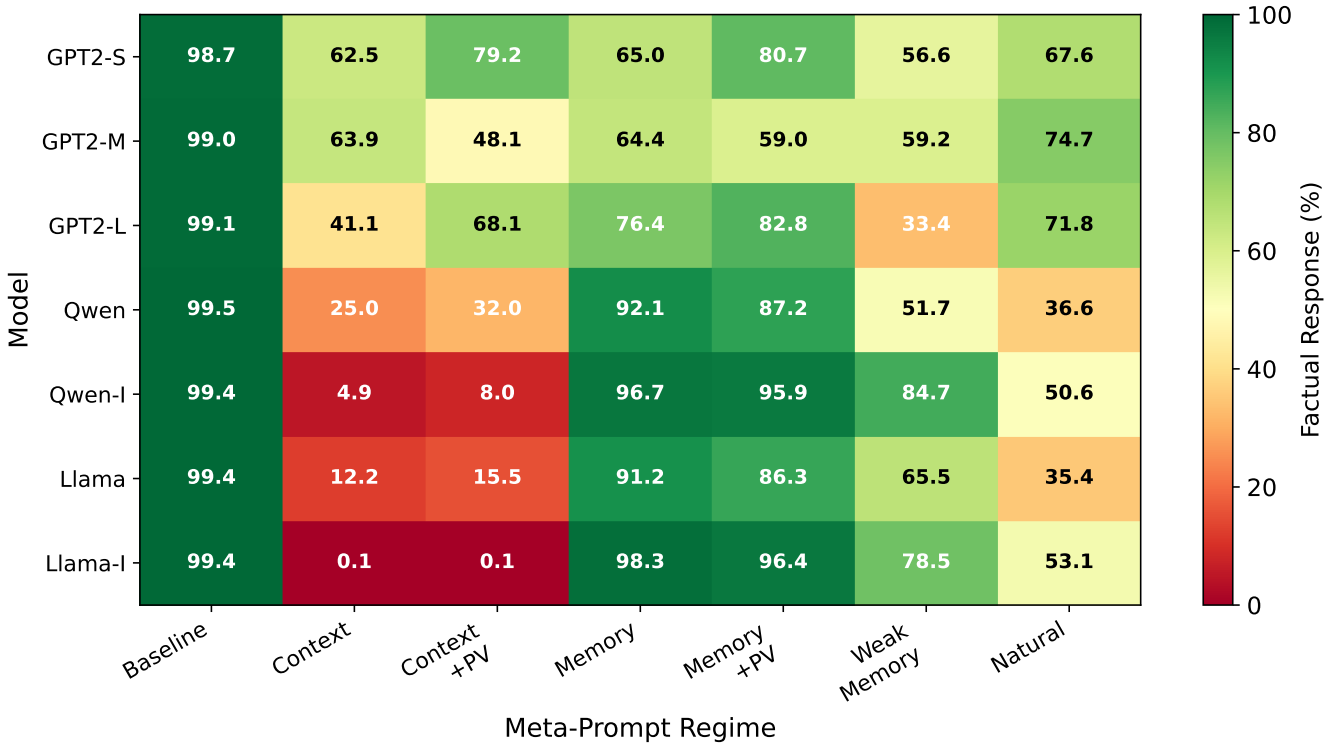


Figure 1: Percentage of factual responses under different meta-prompt regimes across the seven evaluated language models. Lower factual response rates indicate stronger reliance on the supplied counterfactual context, whereas higher rates indicate stronger reliance on parametric memory.

Table 1: Average competition score ($\Delta = \log p(\text{fact}) - \log p(\text{cf})$) under different meta-prompt regimes. Positive values indicate greater confidence in factual knowledge, while negative values indicate greater confidence in the counterfactual context.

Model	Baseline	Context	Context+PV	Memory	Memory+PV	Weak Memory	Natural
GPT2-Small	5.001	0.316	2.160	0.370	2.110	0.110	0.445
GPT2-Medium	6.043	0.328	-0.344	0.402	0.450	0.172	0.952
GPT2-Large	6.038	-0.373	1.823	1.072	2.971	-0.644	0.822
Qwen2.5-3B	8.774	-1.225	-1.042	3.700	3.131	-0.019	-1.352
Qwen2.5-3B-Inst	13.185	-6.806	-5.869	13.247	11.959	5.440	-0.070
Llama-3.1-8B	9.063	-1.153	-1.052	2.599	2.059	0.970	-0.417
Llama-3.1-8B-Inst	9.406	-6.412	-6.023	7.993	6.909	2.405	0.756

score under the Memory regime ($\Delta = 13.25$), while Llama-3.1-8B-Instruct shows a similar shift from $\Delta = -6.41$ to $\Delta = 7.99$. These large confidence changes confirm that instruction tuning induces substantially sharper switching between contextual and parametric reasoning than is apparent from factual response rates alone.

Interestingly, models with comparable factual response rates do not necessarily exhibit similar competition scores. This suggests that answer accuracy alone provides only a partial characterization of context–memory competition, whereas the competition score reveals the strength and stability of the underlying preference between competing knowledge sources.

4.1.6 Scaling versus Instruction Tuning. Figure 2 compares the effects of model scaling and instruction tuning.

Increasing GPT2 model size does not produce a consistent trend in context-memory competition. Although GPT2-Large generally exhibits stronger reliance on factual memory than GPT2-Small under several prompt regimes, GPT2-Medium frequently behaves differently, indicating that scaling alone cannot fully explain competition dynamics.

Instruction tuning, however, consistently transforms model behavior across both Qwen and Llama families. Aligned models exhibit nearly complete switching between contextual reasoning and factual recall depending on prompt instructions.

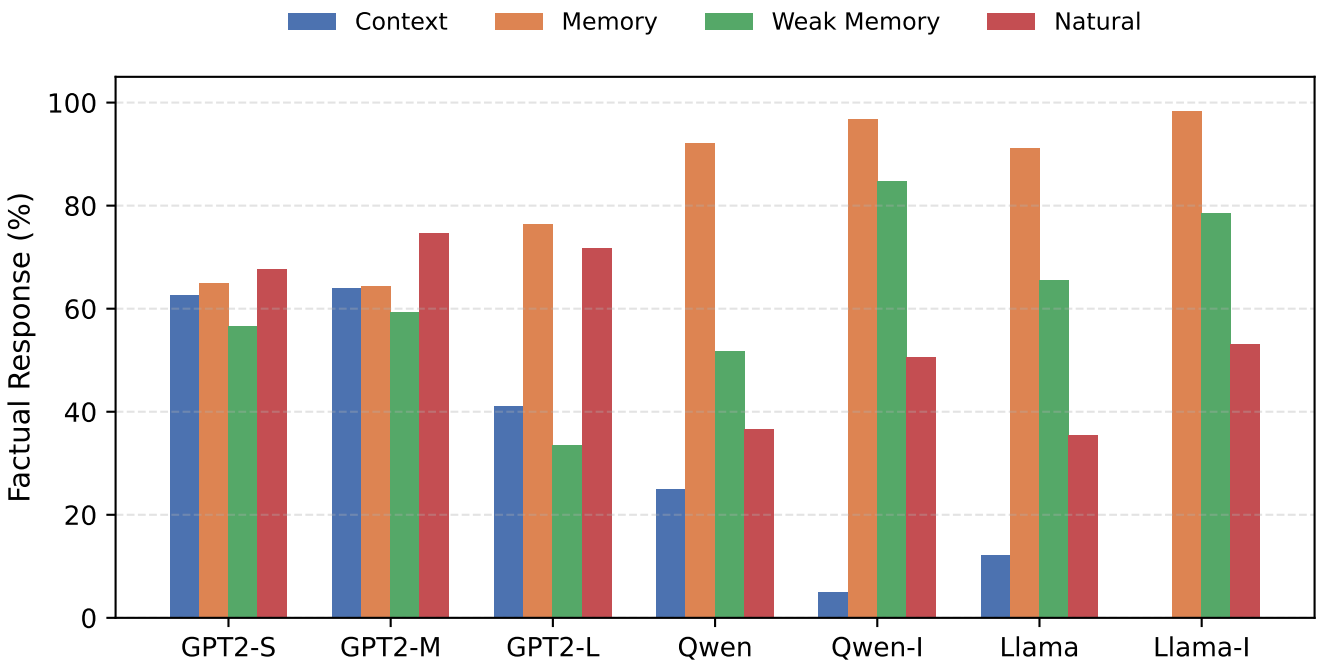


Figure 2: Comparison of representative prompt regimes illustrating the effects of parameter scaling and instruction tuning. Scaling within the GPT2 family produces relatively modest changes, whereas instruction tuning in the Qwen and Llama families results in markedly sharper transitions between contextual and factual reasoning.

These observations suggest that alignment training has a substantially larger effect on context-memory competition than parameter scaling.

4.1.7 Interpretation of Meta-Prompt Competition. Taken together, the results suggest that context–memory competition is governed not only by the information contained in the prompt but also by the extent to which models have been optimized to follow natural-language instructions. The pretrained GPT2 models exhibit relatively gradual transitions across the different prompt regimes, retaining substantial reliance on parametric memory even when explicitly instructed to answer using the supplied counterfactual context. This behavior suggests that contextual information and stored knowledge remains simultaneously active during generation, with neither source completely dominating the other.

In contrast, instruction-tuned Qwen and Llama models exhibit near-binary switching between contextual and factual reasoning. Under context-oriented prompts, these models almost completely adopt the supplied counterfactual information, whereas under memory-oriented prompts they recover nearly perfect factual performance. Such sharp transitions indicate that instruction tuning substantially increases sensitivity to explicit prompt directives, enabling stronger control over which knowledge source is prioritized during generation.

Interestingly, increasing model scale within the GPT2 family does not produce a simple monotonic trend. Although GPT2-Large often shows greater reliance on factual memory than GPT2-Small, GPT2-Medium frequently exhibits intermediate or even divergent behavior across different prompt regimes. These observations suggest that parameter scaling alone is insufficient to explain context–memory competition, and that architectural or optimization differences may interact with prompt formulation in more complex ways.

While our experiments do not directly reveal the internal mechanisms responsible for these behaviors, the observed patterns are consistent with the hypothesis that instruction tuning modifies higher-level policies governing how contextual evidence and parametric knowledge are integrated during inference. Verifying this hypothesis requires mechanistic analysis of transformer activations, attention patterns, and intermediate representations, which we leave for future work.

4.2 RQ2: Effect of Premise-Word Semantics

We next investigate whether lexical framing alone influences competition between contextual information and parametric memory.

Figure 3 presents factual response rates for the five semantic categories. It reveals that context–memory competition is influenced not only by the semantic category of a premise word but also by its specific lexical realization. Although grouped into five semantic categories, the twenty-five premise words exhibit substantial within-category variation. This suggests that language models respond to fine-grained lexical cues rather than relying solely on coarse semantic intent.

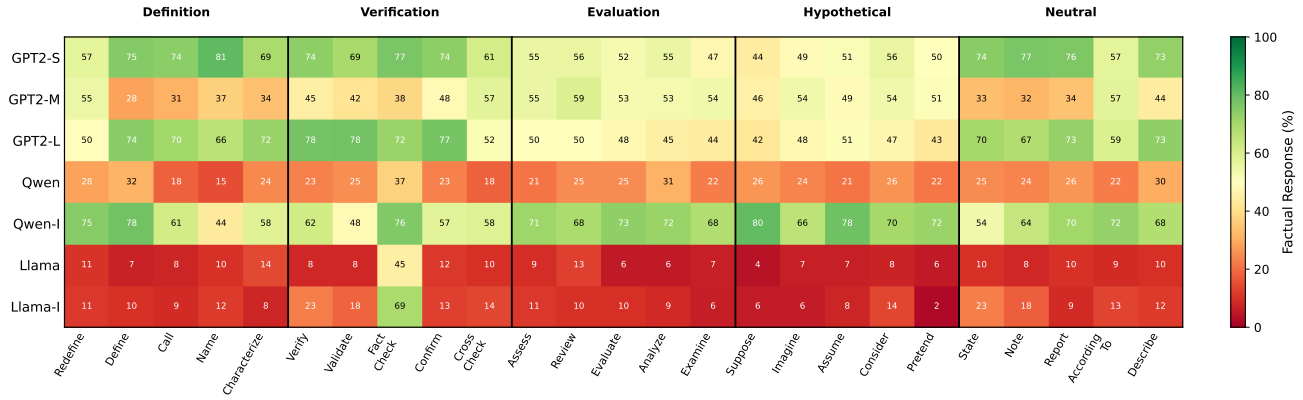


Figure 3: Heatmap of factual response rates (%) under the 25 premise-word prompts. Premise words are grouped into five semantic categories (Definition, Verification, Evaluation, Hypothetical, and Neutral). While category-level trends are visible, substantial variation also exists among individual premise words within the same category, indicating that lexical framing alone can significantly influence context–memory competition.

Within the *Definition* category, for example, premise words such as *define*, *name*, and *call* generally produce stronger factual responses than *redefine*. While all four words are semantically related, *redefine* implicitly signals modification of existing knowledge, making the accompanying counterfactual statement appear more acceptable. In contrast, *define* and *name* are more naturally associated with recalling established facts, encouraging greater reliance on parametric memory.

A similar phenomenon is observed in the *Verification* category. Although *fact check* and *cross-check* both express verification, they produce noticeably different behaviors across several model families. In particular, *fact check* elicits substantially higher factual response rates for both Llama models than the remaining verification terms, suggesting that some lexical formulations are more strongly associated with factual verification than others.

The magnitude of lexical effects also differs across model families. The pretrained GPT2 models exhibit relatively moderate variation across premise words, whereas the instruction-tuned Qwen model shows substantially larger changes between semantic categories. Conversely, the pretrained Llama model remains strongly influenced by contextual information across nearly all premise words, indicating that lexical variation alone is insufficient to overcome contextual dominance. These observations suggest that instruction tuning not only changes the overall balance between contextual information and parametric memory but also increases sensitivity to subtle linguistic framing.

Overall, the results indicate that prompt semantics operate at multiple levels. Broad semantic categories influence model behavior, but individual lexical choices within the same category can further modulate the balance between contextual information and parametric memory. Consequently, prompt engineering should consider not only the intended semantic function of a prompt but also the specific lexical realization used to express that intent.

5 Discussion

Our results extend previous studies of context–memory competition by demonstrating that prompt formulation influences not only the final response selected by a language model but also the confidence with which competing knowledge sources are preferred. While earlier work established that contextual information can override parametric memory, our experiments show that the nature of this competition is strongly shaped by instruction tuning, prompt semantics, and lexical framing.

A key finding is that instruction tuning fundamentally reshapes context–memory competition. The pretrained GPT2 models exhibit relatively gradual transitions between contextual and factual reasoning across the different meta-prompt regimes, suggesting that both contextual information and parametric memory remain active during generation. In contrast, the instruction-tuned Qwen and Llama models display near-binary switching between these two knowledge sources, rapidly changing from context-dominated to memory-dominated behavior when explicit prompt instructions are modified. These observations are consistent with the hypothesis that alignment training strengthens the model's ability to prioritize knowledge sources according to high-level natural language instructions, rather than simply improving instruction following.

Our experiments also demonstrate that prompt semantics operate at a finer granularity than previously recognized. Although the premise words were grouped into five semantic categories, individual lexical choices within the same category frequently produced noticeably different behaviors. For example, semantically related terms such as *define* and *redefine*, or *fact check* and *cross-check*, elicited substantially different factual response rates across several model families. These findings suggest that language models encode lexical priors that influence how counterfactual contexts are interpreted, even when the overall semantic intent of the prompt remains unchanged.

The probabilistic competition score further complements answer accuracy by revealing the confidence with which models prefer

contextual or factual knowledge. Models with similar factual response rates often exhibit markedly different Δ values, indicating that confidence-based analysis provides additional insight beyond discrete predictions alone. In particular, instruction-tuned models show substantially larger shifts in Δ across prompt regimes, highlighting sharper transitions between competing knowledge sources than are evident from accuracy alone.

Collectively, these findings broaden the understanding of context–memory competition beyond its original formulation. Rather than being determined solely by the presence of contextual evidence, competition depends on a combination of prompt structure, lexical framing, model scale, and instruction tuning. From a practical perspective, these observations have implications for prompt engineering, retrieval-augmented generation, and other systems that must balance retrieved context against parametric knowledge. Furthermore, the behavioral patterns identified in this work provide clear hypotheses for future mechanistic interpretability studies using techniques such as activation patching, attention-head analysis, and logit-lens methods to investigate how transformer models internally resolve competition between contextual information and stored factual knowledge.

5.1 Limitations

Our study has several limitations. First, all experiments are conducted on a single counterfactual question-answering benchmark. Although this provides a controlled setting for studying context–memory competition, the observed behaviors may not fully generalize to other tasks such as open-domain question answering, long-context reasoning, or retrieval-augmented generation.

Second, our evaluation is limited to three language model families (GPT2, Qwen2.5, and Llama-3.1). While these models cover both pretrained and instruction-tuned variants, extending the analysis to additional architectures and larger frontier models would help assess the generality of the observed trends.

Third, our analysis is behavioral rather than mechanistic. The competition patterns are inferred from model outputs and log-probability scores, and therefore the proposed explanations for the effects of instruction tuning and lexical framing should be viewed as hypotheses rather than causal conclusions.

Despite these limitations, the controlled experimental design isolates the effects of prompt formulation while keeping the underlying questions and factual knowledge fixed, enabling a systematic comparison of context–memory competition across diverse language model families.

6 Conclusion

This paper investigated how prompt formulation influences competition between contextual information and parametric memory in large language models. Using a controlled counterfactual question-answering framework, we systematically examined two complementary aspects of prompt design: explicit meta-prompt regimes that encourage reliance on contextual information or stored knowledge, and premise-word semantics that vary only the lexical framing of otherwise identical prompts.

Experiments on seven language models spanning the GPT2, Qwen2.5, and Llama-3.1 families demonstrate that prompt design has a profound influence on knowledge selection. Instruction tuning fundamentally changes how models resolve conflicts between contextual information and parametric memory, producing near-complete switching between contextual and factual reasoning under explicit meta-prompts. In contrast, pretrained models exhibit substantially more gradual transitions, suggesting a softer competition between the two knowledge sources.

Beyond explicit prompt instructions, our results show that seemingly minor lexical differences also influence model behavior. Individual premise words within the same semantic category frequently produce markedly different competition patterns, demonstrating that language models are sensitive not only to prompt intent but also to fine-grained lexical realization. These findings highlight prompt semantics as an important yet underexplored factor governing knowledge retrieval in modern language models.

To better characterize this phenomenon, we introduced a probabilistic competition score that measures the relative preference for factual and counterfactual answers. Unlike answer accuracy alone, this metric captures confidence shifts between competing knowledge sources and provides a richer view of model behavior under conflicting evidence.

Overall, our findings suggest that prompt design should be viewed not merely as a technique for improving downstream performance but as a systematic tool for probing and controlling the interaction between contextual information and parametric memory. Understanding this interaction is essential for developing more reliable, interpretable, and controllable language models.

Broader Implications. Understanding how language models balance contextual information and parametric memory has implications for trustworthy AI systems. The observed sensitivity to prompt formulation suggests that seemingly minor linguistic variations can substantially influence model behavior, highlighting the importance of careful prompt design in applications such as retrieval-augmented generation, decision support, and factual question answering.

Future Work. We plan to extend this framework in several directions. First, combining behavioral analysis with mechanistic interpretability techniques [13], such as activation patching, logit lens analysis, and attention-head tracing, may identify where context-memory competition is resolved within transformer architectures. The behavioral trends observed here provide hypotheses that can be tested using activation patching, logit lens, and attention-head analysis. Second, evaluating the proposed framework in retrieval-augmented generation and long-context reasoning settings will help determine whether the observed behaviors generalize to real-world applications. Finally, competition scores may provide useful signals for confidence estimation, hallucination detection, and adaptive retrieval strategies in next-generation LLM systems.

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